Final Exam Formula Sheet

Central Limit theorem:

$$\mu_{\bar{x}} = \mu$$

The standard deviation of the sample means is $\sigma_{\bar{x}} = \sigma / \sqrt{n}$

Confidence interval for μ:

$E = t_{\frac{\alpha}{2}} (\frac{s}{\sqrt{n}})$ if sigma is unknown

$\mu = \bar{x} \pm E$, or $\bar{x} - E < \mu < \bar{x} + E$

Sample size: $n = \left(\frac{Z \times s}{E}\right)^2$

Confidence Interval for π:

$\pi = \hat{p} \pm E$, where

$E = z \sqrt{\frac{\hat{p}\ (1-\hat{p}\ )}{n}}$

Size of sample:

$$n = \frac{z^2 \hat{p}\ (1 - \hat{p}\ )}{E^2}\ if\ \hat{p}\ is\ known$$

$$n = \frac{z^2 (.25)}{E^2}\ if\ \hat{p}\ is\ unknown$$

Test Statistic for Population Proportion

$$z_{test} = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Test Statistic for Population Mean

$$t_{test} = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$$

Two populations:

Formula to test the Difference in Two Dependent (Paired) Populations:

$t_{test} = \frac{\bar{d} - \mu_d}{\left(\frac{s_d}{\sqrt{n}}\right)}$ degree of freedom $= n - 1$

t test statistic (Test the Difference in Means when the variances are different)

$$t_{test} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

df will be given

Test statistic for population proportion difference:

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2} \qquad z = \frac{(\widehat{p_1} - \widehat{p_2}) - (\pi_1 - \pi_2)}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

Correlation test

- Correlation Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad df = n - 2$$